

MORE ROLLS, OR MORE PROGRAMS? SEPARATING EXECUTION SAMPLING FROM TASK SPECIALIZATION IN GENERATED HARNESS POPULATIONS

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ABSTRACT

Automatically generated LLM harnesses can differ in source code and observed outcomes without offering stable task-specific advantages. We study how to separate program-dependent specialization from the benefits of repeated execution, reporting source-code, execution, and outcome diversity as three distinct quantities. In a BIRD text-to-SQL case study, trace inspection finds three execution-level routes through which apparently distinct harnesses fail to realize distinct strategies: claimed mechanisms absent from control flow (6 of 12 generated candidates), mechanisms present but never triggered (`react`, 10/10), and broken multi-turn paths (1/12); code similarity has no rank association with outcome disagreement ($\rho = -0.01$; 21 of 66 generated pairs have identical correctness vectors despite all passing code-difference checks). A separate exploratory study on MATH-500 (35 generated harnesses, 400 tasks, one run each) shows 4.75 pp one-run oracle headroom over the best fixed member (which is the `bare` harness itself). A fresh pre-registered collection then pairs eight generated harnesses against nine byte-identical `bare` clone slots, with three independent repeats each: the repeat-averaged descriptive plugin is positive on both arms despite the clone population having zero specialization by construction (2.54 pp on the real panel’s 350 complete tasks, 2.67 pp on the clone arm’s 383). The cross-fitted frozen-selection gain is $G_{\text{real}} = 0.17$ pp (95% CI $[-0.82, 1.31]$) against $G_{\text{clone}} = 0.55$ pp, so the clone-calibrated difference is -0.10 pp (95% CI $[-1.55, 1.46]$, randomization $p = .7742$) on 342 paired complete tasks; the coverage gate fails (12.5% of real-arm tasks excluded, above the frozen 10% threshold), so the panel reports INSUFFICIENT EVIDENCE — neither a gain nor 1 pp equivalence. Together, these studies show that apparent oracle coverage gains in generated harness populations mix program specialization with the benefits of extra executions of one fixed stochastic program; harness-population evaluation should report one-run coverage opportunity, repeat-stable interaction, and pre-execution selection value as distinct quantities.

1 INTRODUCTION

An LLM harness is the program of prompts, tools, and control flow that wraps a frozen model. When an automatically generated population of such programs shows higher coverage against a best-fixed reference, two explanations are possible: its members may implement distinct task specializations, or the population may simply contain more independent executions of one fixed stochastic program. The two are not the same claim. Two identical programs that each solve a task with probability 0.5 (independent errors) score 0.75 coverage by running both once—a 25 pp gain over running a single one, with zero task-specialization evidence. A population oracle gap can therefore be an observation about execution noise rather than program diversity.

We study this question on BIRD text-to-SQL (Li et al., 2023) and MATH-500 (Lightman et al., 2023), using three sources of evidence. First, in a BIRD discovery population, a trace audit finds that 21 of 66 generated-candidate pairs have identical correctness vectors (36 of 91 including `react` and

bare), even though pairwise code similarity has no rank association with outcome disagreement ($\rho = -0.01$). Three execution-level routes explain the collapse: mechanisms claimed in code but absent from control flow (6 of 12 candidates), mechanisms present but never triggered (`react`, 10/10), and broken multi-turn paths (1/12). Second, a separate exploratory study on MATH-500 (35 generated harnesses, 400 tasks, one run each) shows 4.75 pp one-run oracle headroom over the best fixed member (which is the `bare` harness itself). Third, a fresh pre-registered collection pairs eight generated harnesses (plus `bare`) against nine byte-identical `bare` clone slots, three independent repeats each: the repeat-averaged descriptive plugin is positive on both arms (2.54 pp on the real panel’s 350 complete tasks, 2.67 pp on the clone arm’s 383), yet the cross-fitted frozen-selection gains match those of the same-code arm ($D = -0.10$ pp, 95% CI $[-1.55, 1.46]$, randomization $p = .7742$, on 342 paired complete tasks), and the predeclared coverage gate is unmet on 12.5% of real-arm tasks. Independently, a three-run audit of the single `bare` harness on the 400 BIRD core tasks finds 14.5% of its verdicts changing across repeats, and three `bare` executions treated as three members show a descriptive 6.50 pp one-run gap with no corresponding program specialization.

These observations motivate evaluating apparent complementarity as three separate quantities—source-code diversity, execution-level diversity, and outcome diversity—rather than as a single scalar. Our contributions are:

1. An empirical separation of code-level, execution-level, and outcome-level diversity in generated harness populations, with a trace taxonomy showing exactly where syntactically distinct programs fail to realize distinct strategies.

2. Clone-controlled, repeat-based evaluation. A pre-registered collection with an explicit cross-fitted frozen-selection statistic and a same-code randomization reference measures how much apparent complementarity is attributable to extra executions, with the decision rule validated as instrumentation on known-truth simulation grids.

3. A reporting protocol for harness populations: one-run coverage opportunity, repeat-stable task interaction, and pre-execution selection value reported as separate numbers at separately stated budgets (slots, executions, tokens), with scope stated once per study.

The decisive question is whether distinct generated programs deliver a frozen-selection gain over spending the same executions on one fixed program, with advantages that survive independent repeats. On the fresh pre-registered panel the calibrated difference is indistinguishable from the clone arm’s and the coverage gate is unmet, so the verdict is INSUFFICIENT EVIDENCE rather than either a gain or a 1 pp equivalence: at three repeats, apparent complementarity is consistent with execution sampling rather than a discovered property of the panel. This reframes the target of population construction — useful routing depends on stable task–harness interaction after execution noise is separated, the estimand the controlled study is built to test.

2 WHAT DOES APPARENT COMPLEMENTARITY MEASURE?

An LLM **harness** H is the program of prompts, tools, and control flow wrapped around a frozen target model M : it receives a natural-language task x , issues one or more calls to M , optionally executes intermediate artifacts such as SQL, and returns a final answer scored programmatically. Let H_0 denote the single-call `bare` baseline and $Y(h, x, r) \in \{0, 1\}$ the correctness of member h on task x on independent repeat r . The population P we evaluate always includes H_0 .

Question 1: one-run coverage opportunity. Given a single execution matrix, the *best-fixed accuracy* is $\max_h \mathbb{E}_x Y(h, x, \cdot)$ and the *one-run oracle accuracy* is $\mathbb{E}_x \max_h Y(h, x, \cdot)$. Their difference is the bare-inclusive oracle headroom

$$H(P) = \frac{1}{n} \sum_i \max_{h \in P \cup \{H_0\}} Y(x_i, h) - \max_{h \in P \cup \{H_0\}} \frac{1}{n} \sum_i Y(x_i, h). \quad (1)$$

$H(P)$ records only item-dependent successes present in *that* execution matrix. It cannot establish a repeatable advantage or any pre-execution decision, because it is computed with knowledge of who was correct after the fact. Two simple indicators sit alongside it: pairwise outcome disagreement (the fraction of tasks on which two members differ) and the union repair rate (the fraction of `bare` errors some member corrects); disagreement without repair adds no coverage, and repair that does not beat the best fixed member adds no headroom.

Execution noise is large enough to make the distinction material. On the BIRD core-400 archive, a canonical replay finds 2,391/28,000 same-harness verdict flips against cached outcomes (8.54%), and a dedicated three-run bare-only audit finds 14.5% of tasks changing bare correctness at least once; three independent bare executions treated as three “members” show a descriptive 6.50 pp one-run oracle gap even though they are one program. A single-run $H(P)$ therefore counts both specialization and extra sampling chances, motivating Question 2.

Question 2: repeat-stable task-conditioned advantage. For two members write $\Delta(x) = p_1(x) - p_0(x)$ for their per-task success-probability difference and $(u)_+ = \max(u, 0)$. At the population level,

$$H_{\text{stable}} = \mathbb{E} \Delta(x)_+ - (\mathbb{E} \Delta(x))_+ = \frac{1}{2} (\mathbb{E} |\Delta(x)| - |\mathbb{E} \Delta(x)|), \quad (2)$$

so a member better *everywhere* produces a stable ranking difference with $H_{\text{stable}} = 0$: global dominance is not complementarity, which requires the advantage to cross over tasks. This is a two-member population identity and a conceptual anchor, not a finite-sample bound: mappings and best-fixed members selected from a finite number of noisy repeats carry selection noise, which the same-code control measures in §4.3. Stable ranking and stable complementarity are consequently not a nested ladder.

Question 3: pre-execution selection value. An oracle reads who was correct after execution; a router must commit to one member beforehand, a strictly smaller information set. The practically deployable quantity is therefore the held-out gain of a selector frozen before execution over the best fixed member — not post-execution oracle coverage. The remainder of the paper measures all three questions separately and reports the budgets (slots, executions, tokens) at which each is answered.

3 EVALUATION PROTOCOL

The empirical studies use four evidence sources, presented in the order of §4: a BIRD text-to-SQL discovery population for the code-to-execution trace audit (§4.1); a separate exploratory MATH-500 one-run probe (§4.2); a fresh pre-registered MATH-500 collection with matched repeats and a same-code control (§4.3); and a pre-specified BIRD admission-policy experiment with a versioned corrected reanalysis (§4.4). Each study fixes its own estimand, repeat count, missingness rule, and budget unit, and results are never placed on a before/after line across studies.

Same-code clone control. The decisive design makes “more independent executions of one fixed program” a measured arm rather than a hypothesis. The pre-registered collection runs a panel of eight generated harnesses plus bare and, in parallel, nine equal-slot byte-identical bare clone slots, for three independent repeats each on the same 400 MATH-500 tasks. Equal slots is not equal executions and not equal tokens; all three budget units are reported separately.

Cross-fitted frozen-selection statistic. A repeat-averaged plugin $\hat{H}$ maximizes noisy per-task estimates and is retained only as a descriptive quantity. The scientific statistic cross-fits a frozen selection: in each of three folds, two discovery repeats freeze both the per-task member mapping $h^*(x) = \arg \max_h \bar{q}_h(x)$ and the single best fixed member h_{fix} ; the held-out repeat scores the frozen mapping against that fixed member (never against bare). This gives G_{real} and G_{clone} , and the decision statistic is their paired difference $D = G_{\text{real}} - G_{\text{clone}}$ on tasks complete in both arms, referenced against a two-arm within-task Monte Carlo randomization distribution (member labels permuted independently per task and repeat in both arms) with a paired task bootstrap. The randomization reference is exact at the equal-ability null where arm labels are exchangeable; over the rest of the dominance cone finite-sample false support is checked empirically on known-truth DGPs, not asserted analytically. Frozen safeguards include a $\leq 10\%$ incomplete-task coverage gate in either arm, a ≥ 30 paired-task minimum, and no imputation of unknown cells (all-0/all-1 fill scenarios are reported instead). Without a clone arm the statistic is descriptive and emits no supported complementarity.

Instrumentation validation on known truths. We validate the decision rule on simulated binary tensors with specified member-by-task probabilities, spanning $M \in \{3, 9, 20\}$ members, $T \in$

regime	$R=3$		$R=10$		$R=20$		abst.
	plug	v3	plug	v3	plug	v3	$R=3$
equal $q=0.9^\dagger$	100	0	100	0	100	1	100
equal + difficulty [†]	100	0	100	1	100	1	100
0.5/0.9 dominance [†]	100	0	100	0	13	0	100
dominance [†]	100	0	100	0	100	0	100
cross $H \approx 0.9$ pp	100	1	100	0	100	0	99
cross $H \approx 1.8$ pp	100	1	100	1	100	0	99
cross $H \approx 4.2$ pp	100	0	100	0	100	0	100
two strong specialists, $H=17.5$ pp	100	0	100	100	100	100	100

[†]Null ($H=0$): false-support %, nominal 5%; else power %. $M=9$, $T=400$, $B=300$ at $R=3$ (150 sweeps), 300 randomizations. Wilson intervals and the full M , T , R , missingness grid in Appendix H.

Table 1: **Known-truth instrumentation validation at $M = 9$, $T = 400$.** Null rows report false-support % (nominal 5%), alternative rows report power %; the final column is the $R = 3$ abstention rate. The full M , T , R and missingness grid is in Appendix H.

$\{100, 400, 1000\}$ tasks, $R \in \{3, 5, 10, 20\}$ repeats, complete data, 5% MCAR, and 3% hardness-related missingness, with null families (equal ability, including a heterogeneous-difficulty variant; global dominance, including the 0.5/0.9 configuration) and crossover alternatives of increasing strength. The 100 cells complete with zero procedure errors. Table 1 reports the $M=9, T=400$ slice: the plugin false-supports at rate near one on every null (including all- $q=0.9$ and the sealed clone arm), whereas the clone-calibrated rule’s worst observed null false-support is 2% (equal ability, $M=9, T=100, R=3$; Wilson $[0.68, 5.71]\%$). Power is reported honestly with repeat count: a clean two-specialist crossover at $H = 17.5$ pp is detected with 0% power at $R = 3$, 0.7% at $R = 5$, and 100% at $R = 10, 20$, while realistic 1–4 pp cyclic interactions remain unresolved at $R = 3$. These are tested-DGP statements, not an analytic finite-sample size guarantee over the dominance cone.

Missingness and task provenance. A task is *complete* for an arm if it has a recorded verdict for every member in every repeat; tasks with an unresolved remote call or a carried unknown marker are excluded from that arm rather than imputed. Point estimates are complete-case and exact on the included tasks; the two arms have different complete-task sets, so arm-level descriptive numbers are reported on their own sets and only D is paired across the intersection. The evaluation tasks had been executed in the earlier one-run probe, so the collection is prospective in execution but retrospective in tasks — an independent-repeat study, not unseen-task validation.

4 EMPIRICAL DIAGNOSIS

4.1 CODE DIVERSITY WITHOUT BEHAVIORAL DIVERSITY: BIRD

The discovery population comprises 14 harnesses — 12 evolution-generated candidates plus the `bare` and `react` baselines — with one recorded execution of each harness on each of 60 BIRD tasks (a single-run matrix; matched repeats were not collected here) with a frozen GLM-5.3-Flash target. Pairing its members gives $\binom{12}{2} = 66$ generated-candidate pairs and 91 pairs once the two baselines are included. Every candidate had passed a code-difference check asserting that all files were distinct with non-trivial diffs. Yet 21 of the 66 generated pairs have identical correctness vectors on every task (36 of 91 including the baselines), and across all 91 pairs source-code similarity has essentially no rank association with outcome disagreement (Spearman $\rho = -0.010$; Figure 1). Mean pairwise outcome disagreement is 3.3% overall (3.8% among generated candidates), union repair over `bare` errors is 7.7%, and one-run oracle headroom is 3.3 pp; `react` differs from `bare` on zero tasks.

Auditing execution traces separates three routes through which distinct source code fails to produce distinct executed strategies. **T1 — a mechanism described but not executed** (6/12 generated candidates): the program makes a single model call and runs no artifact, the same execution path as `bare`, while its docstring claims retrieval or self-checking machinery that is absent from the control

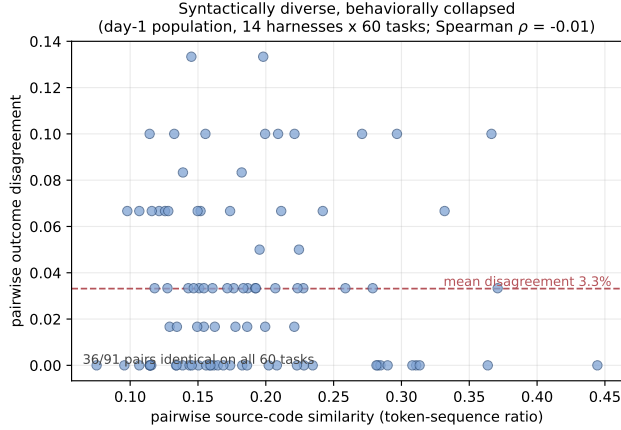


Figure 1: **Code diversity vs. behavioral collapse** (BIRD discovery population: 14 harnesses $\times$ 60 tasks, 91 pairs including `bare` and `react`). Each point is a harness pair; x is token-sequence code similarity and y the fraction of tasks with differing correctness. Code similarity does not predict outcome disagreement ($\rho = -0.010$), and 36/91 pairs are identical on all 60 tasks.

flow. **T2 — a mechanism present but never triggered** (`react` on 10/10 audited tasks): the error-repair branch is real code, but the first SQL succeeds on every audited task, so the branch never fires and the final SQL equals `bare`; this does not show the repair path lacks causal force under failure. **T3 — a broken multi-turn path** (1/12): an execute-inspect-repair program returns the model’s prose as the final SQL, turning `bare`-correct items into harness-wrong ones. The three classes ask for different tests — execution-path acceptance, controlled fault injection, and executable-output validation, respectively.

A human positive control shows the apparatus can detect genuine specialization. Four hand-written harnesses with forcibly distinct execution paths (execute-and-repair, three-vote, schema-linking, and hint-first reordering) under the same frozen target and judge give oracle headroom 6.67 pp on the 60 tasks and 7.28 pp (95% interval [3.3, 10.0]) on the full 151-item set, with each mechanism repairing tasks the others miss. This control was authored after inspecting the discovery traces, so its headroom is descriptive rather than blind; the discovery population’s provenance, including the 24/151 items overlapping protocol development and the conservative proxy judge, is reported in Appendix A.

4.2 ONE-RUN HEADROOM IN AN EXPLORATORY MATH-500 POPULATION

A separate exploratory study on MATH-500 uses six builders, one seed, a mechanism-blind validity gate, and the frozen GLM-5.3-Flash target at temperature 0, with one execution of each admitted harness on the same 400 evaluation tasks. Of 48 generation slots 35 admitted; two reasoning-heavy builders exhausted the 32k generation budget on most slots, so the population under-represents reasoning-heavy styles. This is a replication probe, not a factorial, and it collected neither repeats nor a same-code control.

The population looks behaviorally diverse: mean pairwise outcome disagreement is 10.6% (maximum 33.0%), 33 of the 36 members (counting `bare`) have unique correctness vectors, and members repair 76.0% of `bare` errors. But one-run oracle coverage is 98.50% while the best *fixed* member is `bare` itself at 93.75%; every generated member is individually worse (0.6825–0.915). The `bare`-inclusive one-run oracle headroom is thus 4.75 pp: a coverage opportunity that cannot be realized by committing to any single program before execution. This exploratory observation motivated the separate pre-registered repeat-and-clone study of §4.3; the two studies use different populations, estimands, and budgets, and the 4.75 pp is not compared with the controlled result as a before-and-after correction. The second-domain replication is detailed in Appendix J.

Quantity	Real arm	Clone arm
complete tasks	350	383
repeat-averaged plugin $\hat{H}$	2.54 pp	2.67 pp
cross-fitted gain G	0.17 pp	0.55 pp
calibrated $D = -0.10$ pp, CI $[-1.55, 1.46]$, $p = .7742$		
paired complete tasks: 342; exclusions 12.5%/4.2%		

Table 2: **Clone-controlled MATH-500 results.** The arm-level plugin and G use each arm’s own complete-task set (350 real / 383 clone), so neither they nor their arithmetic difference are a paired contrast; D recomputes both gains on the 342 tasks complete in both arms, which is why $D \neq G_{\text{real}} - G_{\text{clone}}$ as printed in the arm rows.

4.3 SAME-CODE CLONES SHOW POSITIVE APPARENT COMPLEMENTARITY

The pre-registered collection (frozen 2026-09-15) executes a panel of eight generated harnesses plus bare against nine *equal-slot* byte-identical bare clone slots, for three independent repeats each on the same 400 MATH-500 tasks. The panel is a salted-hash draw from the 35 exploratory members, with eligibility restricted to low-call mechanisms; the evaluation tasks had been used in the one-run probe, so this is a prospective-execution, retrospective-task study rather than unseen-task validation. Protocol and execution records are in Appendix I.

The repeat-averaged descriptive plugin is positive on *both* arms — 2.54 pp on the real panel’s 350 complete tasks and 2.67 pp on the clone arm’s 383 — even though the clone population has $H_{\text{stable}} = 0$ by construction. This is the paper’s sharpest measurement fact: same source code, same repeat schedule, positive apparent complementarity, exactly as the known-truth nulls predict for a maximization-based plugin.

The cross-fitted frozen-selection gain is $G_{\text{real}} = 0.17$ pp (CI $[-0.82, 1.31]$) and $G_{\text{clone}} = 0.55$ pp on the two arms’ own complete-task sets, so the finite- (R, T) selection noise on the clone arm is of the same order as the panel’s gain. The decision statistic D is not the arithmetic difference of those arm-level values: both gains are recomputed by cross-fitting on the 342 tasks complete in both arms, giving $D = -0.10$ pp (CI $[-1.55, 1.46]$, randomization $p = .7742$). Stable member ranking (C-rank) is SUPPORTED, but the complementarity decision abstains: 12.5% of real-arm tasks are incomplete, above the frozen 10% coverage gate, so C-comp is INSUFFICIENT EVIDENCE, the interval’s upper end does not establish equivalence at the 1 pp margin, and the all-0/all-1 fill scenarios for excluded tasks bracket G at $[-0.33, 0.33]$ pp. The abstention is power-driven as well as coverage-driven: at $R = 3$ the instrument detects even a strong $H = 17.5$ pp two-specialist crossover with 0% power (0.7% at $R = 5$, 100% at $R = 10, 20$; §3).

Slots do not determine spend. Offline common execution-budget replay matches post-execution oracle coverage of the dev-ranked panel against b of the 27 same-code executions at $b \in \{3, 9, 15, 27\}$: the panel minus same-code differences are +0.37, +0.03, −0.11, −0.25 pp, i.e. tied at every budget (range −0.25 to +0.37 pp). The panel used 3.73× the clone arm’s tokens (54.17M vs 14.51M, 22,192 vs 10,800 logical calls), so no realized cost-matched claim is made, and no cross-arm majority-vote accuracy is reported because the arms emit incompatible answer formats. The frozen pre-execution selector π_Z abstains to bare on every eval task and ties dev-fixed at 94.96% over the three repeats (the single-run bare accuracy in the exploratory study was 93.75%; the higher value here is the repeat-mean on the same task set, a different statistic, not a contradiction), while a pre-fixed random member is materially worse (−10.12 pp, CI $[-13.27, -7.22]$): the observed oracle value is not realized routing value.

4.4 AN ADMISSION-POLICY STUDY ON BIRD

A separate confirmatory BIRD study crosses six independent builders with three seeds under strategy forcing (a named mechanism from a fixed set versus free invention) and conformance gating, giving four arms: II-A (free, ungated), II-B (free, gated), II-C (forced, ungated), and II-D (forced, gated). The evaluation covers 1,169 questions from nine databases never opened during development (one disclosed 18-item smoke-test exception), with the official BIRD execution scorer and a pre-outcome-

Metric	II-A	II-D
Candidate count	5.78	3.83
Oracle accuracy (%)	76.14	75.70
Best-fixed accuracy (%)	67.58	67.30
Headroom (pp)	8.56	8.40
D–A headroom (pp)	–0.16	
Unadjusted bootstrap 95% CI	[–0.83, +0.87]	
One-sided sign-flip p	0.647	

Table 3: **Corrected primary analysis.** Means over 18 builder–seed cells on 1,169 questions; bare is included in both the oracle and best-fixed comparators.

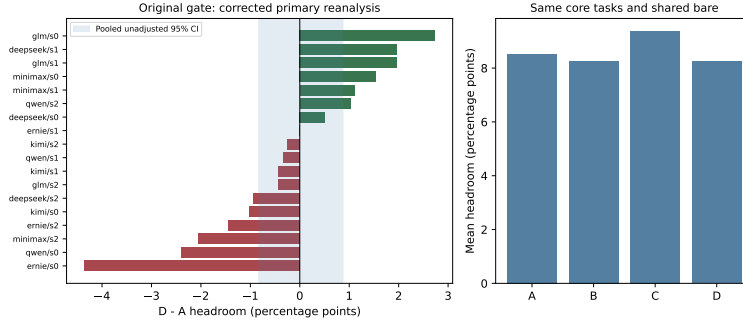


Figure 2: **Versioned corrected reanalysis.** Left: full-set D–A cell contrasts with the pooled 95% interval. Right: four-arm headroom on the shared core-400 tasks with a common bare baseline.

frozen analysis later re-run as a versioned corrected reanalysis. The implemented gate bundles nontrivial-mechanism structural screening with conformance verification — two of eight forced prompt-transformation slots cannot pass even when neutrally valid — so the treatment does not isolate verification causally.

The diagonal forced/gated minus free/ungated contrast in bare-inclusive oracle headroom is -0.16 pp (unadjusted 95% CI $[-0.83, +0.87]$, one-sided sign-flip $p = 0.647$), with cell contrasts ranging over -4.36 to $+2.74$ pp: the bundled screening policy shows no detected headroom dividend, and the interval does not establish equivalence.

On the shared core-400 tasks the gate contrast is -0.69 pp (Holm-adjusted $p = 0.079$), and the strategy and interaction secondaries are likewise inconclusive (Table 4); the fixed- K sensitivity -0.31 pp (CI $[-1.07, +0.40]$) confirms headroom is non-monotone in population size. The open-mechanism arm ran only on the development side and has no confirmatory estimand. These are all single-run contrasts, and the execution variability reported in §2 (8.54% R2 verdict flips; 14.5% of tasks changing bare correctness across three runs; a 6.50 pp same-code one-run gap) characterizes them as coverage opportunities rather than capability differences.

5 RELATED WORK

5.1 HARNESS GENERATION FOR LLMs

A growing line of work generates task-specific scaffolding around a frozen model. TTHe evolves text-to-SQL harnesses with an LLM proposer/judge loop (Nie et al., 2026); Self-Harness (Zhang et al., 2026b) shows a model improving its own harness; JIT-Agent (Zhang et al., 2026a) learns to generate a harness per task at inference time; and Task-CoEvolve (Miyai et al., 2026) co-evolves tasks and scaffolds. Automated agent-program search (Hu et al., 2024) first framed such generated programs as a search space. A concurrent 2026 cluster makes harness evolution behavior-aware from the verification side: HarnessLens (Xu et al., 2026) allocates rollout budget using behavior-level checks during evolution, HarnessFix (Chen et al., 2026a) repairs harnesses from trace-grounded

Contrast	Effect (pp)	95% CI (pp)	Raw p	Holm p
Gate	-0.69	$[-1.56, +0.20]$	0.026	0.079
Strategy	+0.42	$[-0.32, +1.51]$	0.263	0.428
Interaction	-0.87	$[-2.83, +0.65]$	0.214	0.428

Table 4: **Corrected core-400 factorial reanalysis.** Gate is $[(B-A) + (D-C)]/2$, strategy $[(C-A) + (D-B)]/2$, interaction $(D-C) - (B-A)$; raw and Holm-adjusted p -values.

	This work	HarnessLens	HarnessBank
Unit of study	population outcome matrix	evolution loop	diversity library
Behavioral collapse measured	yes	no	search collapse only
Same-code clone control	yes	no	no
Admission check	counterfactual traces	budget allocation	semantic screening
Policy comparison	factorial	single loop	none
Confirmatory split	9 isolated DBs	—	—

Table 5: Nearest concurrent systems. The present gate includes structural screening, and the confirmatory split has a disclosed 18-item smoke-test exception.

diagnoses, AHE (Lin et al., 2026) verifies edits against self-declared behavioral predictions, gated quality-diversity/HarnessBank (Luo et al., 2026) evolves populations over LLM-assigned pathology descriptors, the Harness Handbook (Wang et al., 2026) targets behavior localization and outer-loop search, and HarnessForge (Chen et al., 2026b) co-evolves the harness and the internal policy.

Our unit of analysis and question are complementary but distinct. Table 5 contrasts the two nearest systems. HarnessLens uses behavior as a *signal inside* an evolution loop to allocate compute; we ask whether the resulting *population* is behaviorally diverse as an outcome property. HarnessBank (Luo et al., 2026) documents search-space collapse and screens on LLM-assigned semantic descriptors; our admission instrument does not score semantics at all, but drives counterfactual trace probes to verify mechanically that a declared mechanism executes, and our experiment compares admission policies through a factorial contrast. Because the implemented gate also structurally excludes two allowed prompt transformations, the treatment does not isolate verification’s causal contribution.

5.2 CONDITIONAL UTILITY, ROUTING, AND REPEATED SAMPLING

Harness quality is instance-conditional: Gupta et al. (2026) and Cho (2026) document that no single harness dominates across instances. HELIX evaluates a full sibling-harness outcome matrix and portfolio complementarity beyond the best fixed candidate but lists a router as future work (Fan & Huang, 2026). STS routes over six fixed human-designed scaffolds (Zhou et al., 2026), GRASP gates skill edits into a development-time library (Moll et al., 2026), Kim (2026) learn routing over {no-search, search, abstain}, and RouteLLM (Ong et al., 2024) routes at the model level. The underlying decision boundary is the classical distinction between quantitative and qualitative (crossover) interaction: a change in effect size alone need not change the preferred action (Gail & Simon, 1985). Algorithm selection (Rice, 1976) and portfolio solvers such as SATzilla (Xu et al., 2008; Kotthoff, 2014) select enumerated algorithms from instance features, and randomized restarts have long captured gains from additional attempts at one solver rather than from a portfolio. Ensemble selection optimizes a validation objective over a model library (Caruana et al., 2004), a different information set from choosing one member before execution, and self-consistency (Wang et al., 2022) aggregates multiple sampled trajectories post hoc, so its cost and information set again differ from pre-execution single-member routing.

Our candidates are complete automatically generated harnesses, and we measure the population outcome structure a selector would inherit rather than build a selector. Post-execution verifiers (Kwok et al., 2026) score completed trajectories, whereas our pre-execution estimand chooses a harness before it runs, a different information set; our oracle coverage is labeled throughout as the post-execution upper bound a verifier could realize, not as routing utility.

5.3 BEHAVIORAL DIVERSITY IN GENERATED POPULATIONS

Diversity of LLM outputs has been studied mainly at the sample level (pass@ k (Chen et al., 2021), self-consistency, diverse decoding) or at the agent-policy level. Repeated sampling from a fixed generator and multi-agent ensembling of independently instantiated identical agents (Li et al., 2024) already show performance gains from additional executions of the same code, and nominally temperature-0 calls vary across repeats (Alizadeh et al., 2023); our same-code clone arm turns that execution-level baseline into an explicit measured null for harness populations. Syntactically distinct programs behaving identically is adjacent to classical program-equivalence (Budd & Angluin, 1982) and equivalent-mutant (Papadakis et al., 2019) results. Most closely, El et al. (2025) report low behavioral diversity among auto-designed agents at an aggregate level. We extend that observation in two directions: a trace-level taxonomy (mechanism described but not executed, present but untriggered, broken multi-turn) showing exactly where syntactic difference fails to become an executed strategy, and the converse finding that identical code under repeated execution produces a positive apparent-complementarity signal — measured, for harness populations, with matched repeats, a same-code control, and explicit slots/executions/tokens bookkeeping.

6 IMPLICATIONS AND LIMITATIONS

What the studies show. Population utility is a stable task–harness interaction after execution noise is separated: an oracle asks who was correct after execution, whereas a router must commit beforehand. On BIRD, source-code diversity is uninformative about behavioral difference ($\rho = -0.01$, 21/66 identical vectors despite passing code-difference checks), with three named execution routes showing where distinct programs fail to realize a distinct strategy. On the exploratory MATH-500 population, one-run oracle coverage is 4.75 pp above the best fixed member, which is `bare` itself. On the pre-registered panel, byte-identical programs reproduce a positive repeat-averaged plugin (2.67 pp on the clone arm), while the calibrated difference is $D = -0.10$ pp (CI $[-1.55, 1.46]$, $p = .7742$) and common-execution coverage is tied: at three repeats the apparent gain is indistinguishable from extra executions of one fixed program, and the predeclared rule abstains rather than calling this equivalence; the frozen selector captures none of the gap a pre-fixed random member loses (-10.12 pp).

A reporting protocol. The results motivate four low-cost practices: (i) include a same-code clone arm so additional executions are a measured baseline; (ii) collect held-out independent repeats with discovery/validation splitting, and report the repeat count and per-arm missingness; (iii) report one-run coverage opportunity, repeat-stable interaction, and pre-execution selection value as three distinct numbers with distinct information sets; (iv) report slots, executions, and tokens separately, running any budget-matched comparison at the claimed unit.

Scope. The BIRD evidence covers one SQL target, six builders, and three seeds (W1/W3 analyses descriptive); the MATH panel is a low-call-filtered draw on previously seen tasks, not an unseen-task test. At $R = 3$ the decision is powered only for strong crossovers (0% power for an $H = 17.5$ pp interaction) and the coverage gate independently failed on 12.5% of real-arm tasks, so the abstention is a measurement boundary, not evidence of absence — more repeats are the direct route to resolving it. The panel spent $3.73\times$ the clone arm’s tokens, so the equal-execution replay (not realized cost) is the cost-controlled comparison, and calibration is an empirical tested-DGP result over 100 cells, not an analytic dominance-cone guarantee. The portable claim is the measurement distinction and its controls across two domains; other panels, targets, and repeat counts are the open extension.

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ETHICS STATEMENT

All experiments use public BIRD and MATH-500 benchmark data and generated program artifacts; no human participants, private records, or personally identifiable information are collected. Model calls are reported as part of the experimental provenance, and the paper does not release credentials

or provider responses that would expose private data. The main ethical risk is overstating what one-run oracle gains imply for deployed routing; we therefore report the unresolved selection and cost claims explicitly as INSUFFICIENT EVIDENCE.

REPRODUCIBILITY STATEMENT

The supplementary source includes the frozen diagnostic implementation, control manifests, and hash-bound result records. ‘REPRODUCE.md’ gives the exact offline commands for unit tests, calibration, blinded controls, archive diagnoses, and the LaTeX build. The reported provider-dependent archives are retained with their provenance and hashes; no claim requires access to a live model API to rerun the diagnostic itself.

AI USE STATEMENT

This work is, by construction, an AI-assisted research project studying AI-generated artifacts, and we disclose the extent of that assistance precisely.

Required disclosure. We used generative AI tools for: (i) writing and modifying the experimental code (generation pipeline, evaluation collectors, statistical analysis scripts); (ii) running experimental workflows that made model API calls; (iii) analyzing experimental data and producing tables and figures; and (iv) drafting and editing manuscript text. The revised primary and factorial tables and result figure are generated from a shared, versioned reanalysis artifact. Historical exploratory analyses retain their original provenance and stated limitations. The post-review corrections are recorded separately from the original protocol. Verification records specify the scope of automated tests, development controls, and review for each artifact. Post-review revisions also used same-model subagent reviews; these are mechanical checks, not independent cross-model scientific validation. Development controls and local runtime tests do not establish independent gate calibration or provider-level execution independence.

Research ideas and direction. The core research questions, the causal design decisions (estimand, contrasts, exclusion rules), the interpretation standards, and proposed go/no-go criteria were developed with AI assistance. The human authors set the overall research direction and requested the review-driven revision; AI tools also proposed analysis and experiment designs, identified counterexamples, and implemented revisions. Final scientific decisions and approval of the submission remain the authors’ responsibility. This disclosure does not claim that every AI-proposed decision has already received individual author approval.

Responsibility. We take responsibility for the final content of this work, including text, claims, and artifacts produced with the aid of generative AI tools. The factual statements in this section are provided for author verification; final responsibility for their accuracy rests with the authors.

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A PHASE-I PROVENANCE AND CONDITIONS (DETAIL)

Discovery status. Phase I (§4.1 and C.1) is a discovery study. 24 of its 151 evaluation items overlapped day-1 protocol-development measurements; the human control was designed after inspecting day-1 traces (its observed headroom is descriptive, not a blind estimate of stable complementarity); the repository history was initialized retrospectively, and the decision-log changelog is the authoritative ordering record. Phase-I admission thresholds (union repair $\geq 15\%$, headroom ≥ 5 pp, ≥ 3 pairwise-distinct non-empty fix sets) were frozen in the internal decision log before the 151-item evaluation; the v6→v7 threshold change (20%→15%) is disclosed, and both readings admit the same populations. This is an internal pre-evaluation freeze, not external preregistration.

Data hygiene. The bare baseline flipped 3 of 60 items between the two day-1 runs (API nondeterminism at temperature 0), so 60-item columns are conditioned on slightly different bare-error sets. Phase-I judging used a set-of-rows-equality proxy that we later measured to be uniformly conservative relative to the official BIRD scorer (on the two Phase-I matrices with logged final SQL, it never credits an answer the official scorer rejects; agreement 0.97–0.99).

Phase-I populations. Seed baselines: *bare* (single call) and *react* (execution feedback loop, shipped with TTHE). *Old proposer protocol*: six candidates selected from TTHE’s optimize-loop output (GLM-5.3-Flash builder), selected for behavioral diversity using only the 36 questions disjoint from the evaluation set, by a rule frozen before evaluation. *Strategy-forced free-form*: GLM-5.3-Flash, Qwen3.8-Flash, and DeepSeek-V4-Flash-Vision-Exp builders, eight assigned mechanisms each; the GLM population is the logged re-run ($K=3$; the original 09-01 run recorded zero accepted outputs but its tool was not preserved and the observation is unaudited). *Harness IR*: six harnesses compiled deterministically from declarative mechanism specifications (context $\in \{\text{plain, schema.link, hint.guard, link+hint}\}$, candidate generation with temperature, per-candidate execution, repair rounds, selection rule), so a declared mechanism is present in the control flow by construction. *Human positive control*: the four hand-written harnesses of §4.1.

Generation-provenance gaps. The 09-01 free-form populations were produced in-session; their generation tool was not preserved, and per-attempt logs do not exist for them — so budget equality cannot be audited for those runs, and we do not claim it. Their acceptance was an import/smoke check at evaluation time, *not* the Phase-II counterfactual conformance gate. The tool of record from 2026-09-03 (`builder_generate3.py`) logs every attempt (3 retries, 16,384-token budget, per-attempt stage and finish reason); the completed per-strategy grids and the builder-acceptance ordering (3/8, 7/8, 8/8) come from this logged tool and are post-hoc with respect to the frozen admission decisions.

A.1 PHASE-I PROTOCOL PROBES AND THEIR LIMITS

Phase I also probed generation-side fixes — a strategy-forced free-form protocol (assigned mechanism, import/smoke acceptance) and a compiled Harness IR. Under the strategy-forced protocol two of three builders produced populations whose point estimates exceeded the pre-specified thresholds (Qwen3.8-Flash 5.96, DeepSeek-V4-Flash-Vision-Exp 7.95 pp; the old-protocol arm missed at 4.64 pp), and per-strategy acceptance ordered with builder identity (GLM-5.3-Flash 3/8, Qwen3.8-Flash 7/8, DeepSeek-V4-Flash-Vision-Exp 8/8, under the logged tool). These observations motivate Phase II but are not confirmatory, for three disclosed reasons: the design was two nested contrasts that cannot separate protocol from builder main effects; the early runs lack per-attempt logs, so budget equality cannot be audited; and acceptance was an import/smoke check, not the counterfactual conformance gate of §3. Full condition details are in Appendix A.

Generation-null failures (M0). Re-running the original optimize loop with Qwen3.8-Flash and DeepSeek-V4-Flash-Vision-Exp builders produced 40 candidates byte-identical to the `bare` baseline source. Code identity alone does not imply identical stochastic executions. This run supplies no distinct-code population; it is reported as a distinct, more primitive failure mode — the pipeline produced no distinct population at all. The collapse claims above rest exclusively on the genuinely distinct-source day-1 population.

Figure data. The raw pairs behind Figure 1 (in the main text) are `fig1_phenomenon_data.json` in the code release.

B PHASE-II PROTOCOL SUMMARY AND DEVIATION LOG

The authoritative protocol (estimands, splits, instruments, exclusion rules, and the historical deviation changelog) is `PHASE2_PROTOCOL.md` in the code release, frozen at the commit recorded in `PROTOCOL_FREEZE.txt`. Summary: 2×2 factorial plus open-mechanism arm; six builders × three seeds; at most $R=3$ attempts per slot, stopping at first admission, with no top-up; contamination registry (fail-closed, two development databases banned wholesale); 1,169-question confirmatory set from nine databases never opened during development plus a pre-drawn 400-item core; official BIRD judge with enforced 30 s timeout; response cache keyed on model identity with cross-process locking (deviation D13 documents and fixes a cache concurrency bug found during collection); primary contrast D—A on paired (builder × seed) cells with hierarchical bootstrap and permutation inference; Holm-corrected factorial secondaries on the common core; K-matched sensitivity only. Deviations D1–D13 include the disclosed 18-item pipeline smoke test, instrument-calibration fixes made before any confirmatory outcome was observed, the credential-handling incident and its remediation, and the artifact-divergence quarantine — each with its rationale and date in the changelog. The post-review corrections are recorded in `review-stage/REVISION_20260910.md`; revised results do not replace the historical freeze. In particular, the implemented gate structurally rejects two allowed prompt-level strategies and realized generation costs differ across arms. Corrected tables use shared baseline records and database/task resampling, as described in the main text.

Upstream code-extraction audit. Across the 36 forced C/D cells, all 108 `format_guard` attempts failed neutral validity: 71 logs report unterminated Python strings, 36 report no Python fence, and one reports a provider timeout. The archived generator’s non-greedy Python-fence extractor terminates at the first subsequent triple backtick, including one inside a Python string. Three synthetic, syntactically valid files containing SQL fences in an ordinary string, docstring, or regex

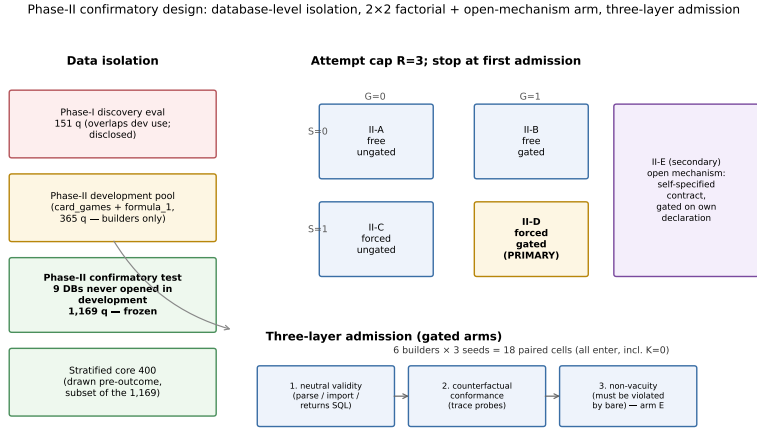


Figure 3: **Phase-II confirmatory design** (frozen protocol). Left: database-level contamination isolation. Middle: the 2x2 factorial with a common attempt cap, plus open-mechanism arm II-E. Right: the three-layer admission pipeline.

become invalid prefixes; the extractor nevertheless sets its truncation flag to false. This extractor is unchanged from the review-locked commit, but the historical attempt logs do not bind its execution-time source hash. Full builder messages were not saved by this generator. Thus the failure pattern is compatible with extraction damage, not proof that all historical failures had this cause or that repaired candidates would pass admission. We found 74 saved extracted files, all syntactically invalid; three coexist with no-fence logs and are retained as provenance conflicts, not attributed to those attempts. These findings place a generation interface failure alongside the gate’s explicit strategy exclusions.

Corrected K-matched and R2 replays. Both replays use the same manifest-ordered canonical loader and shared database/within-database task draws as the corrected primary analysis, with seeds resampled within fixed builders (10,000 replicates). For K-matching, each A/B and C/D pair uses the smaller candidate count; every uniform candidate subset is enumerated exactly. Bare stays in every subset, and best-fixed is recomputed inside each subset on every task draw. This is a size sensitivity analysis, not a causal mediation decomposition. Retaining all 18 cells, including two with singleton A subsets, changes subset D—A from +0.42 to +0.81 pp (change +0.39 pp, 95% CI [-0.95, +1.63]). R2 subset headroom excludes bare because R2 did not acquire a matched bare run, so it is distinct from the bare-inclusive primary estimand; the old 16-cell comparison is reconciled here. These are instability measurements, not standard errors or a detectability floor for the mean treatment effect; they do not establish that every contrast is unchanged, or identify what fraction of between-harness disagreement is noise.

R2 acquired 70 harnesses on 400 tasks, selecting the first two callable admitted candidates per A/D cell where available. The two cache-off acquisition files retain separate identities even though both stored `repeat=0`; source hashes are checked against the cached runs. No matched bare was acquired, so R2 headroom excludes bare in both conditions. The corrected analysis retains singleton cells (their subset headroom is zero), yielding all 18 paired cells. There are 2391/28,000 verdict flips against cached outcomes (8.54%) and 1245/14,400 flips between the two D cache-off passes (8.65%). For reconciliation only, restricting to the 16 cells with at least two members in both arms gives D—A -0.11 pp cached and +0.36 pp cache-off. This explains the difference from the full-cell estimates; neither analysis identifies future execution variance, the effect of caching alone, or a stable-complementarity advantage over clones. Acquisition time and caching condition are not isolated.

R3 bare-only repeat audit. R3 is a separate, smaller instability audit: three independent cache-off executions of the *bare baseline only* (no generated harnesses) on the same core-400 tasks, same frozen target (GLM-5.3-Flash, temperature 0). 58 of 400 tasks (14.5%) changed bare correctness in at least one of the three repeats; pairwise flip rates between re-

Sensitivity estimand	Estimate (pp)	95% CI (pp)
K-matched B–A headroom	−0.55	[−1.53, +0.63]
K-matched D–C headroom	−0.08	[−1.26, +0.80]
K-matched gate mean	−0.31	[−1.07, +0.40]
R2 subset D–A (cached)	+0.42	[−0.61, +1.48]
R2 subset D–A (cache-off)	+0.81	[−0.35, +1.84]
R2 change in subset D–A	+0.39	[−0.95, +1.63]

Table 6: Corrected sensitivities on core-400, 18 cells. CIs are unadjusted. K-matched comparisons always retain bare; R2 comparisons exclude bare and are conditional on the observed acquisition groups.

Arm	Candidate-task records	Logged solver calls	Calls/task/population
A	41,600	67,258	9.34
B	44,000	64,399	8.94
C	40,400	75,604	10.50
D	27,600	52,043	7.23

Table 7: Descriptive canonical cost audit on core-400, 18 cells per arm. Logged logical calls are not billable requests or token counts. The last column averages candidate-population calls per task over cells.

peat pairs are 39/400, 40/400, and 37/400 (9.75%, 10.0%, 9.25%). Treating the three bare repeats as three pseudo-members (one execution each) gives a one-run oracle of 72.25% against a best-single-repeat accuracy of 65.75%, a descriptive 6.50 pp same-code oracle gap computed from the raw `artifacts/phase2/r3_rep{0,1,2}.jsonl` official-scorer columns. R3 therefore measures the observed instability rate of the baseline itself under temperature-0 API nondeterminism. This is not a standard error or an effect-size floor for any contrast, but it sets the scale at which repeat controls become necessary for interpretation: a harness-versus-bare difference of this magnitude cannot be attributed to capability without them. The archived record is `artifacts/phase2/r3_flip_rate.json` with the generating script `experiment/phase2/run_r2r3.py`.

Recorded cost and missing billing data. The same canonical core records contain 259,304 logged logical solver calls across 153,600 candidate-task records (Table 7). These calls include cache-served requests and are not provider-billed API requests. None of the inspected records has token usage fields; we do not impute token or monetary expenditure from call counts. Actual generation attempts, admitted candidate counts, and these execution counts therefore remain separate measurements. The population cost mean gives each builder-seed cell equal weight and excludes bare; zero-candidate populations would contribute zero solver calls.

B.1 POST-REVIEW PROVIDER ACQUISITION STATUS

The post-review provider acquisition attempts did not yield a complete common pool: transport interruptions left completed responses, unclosed requests, and unstarted attempts in separate auditable states. These records are retained for provenance and missingness accounting, but are excluded from claims about population performance, cost, or stable complementarity. Thus, the manuscript does not use provider availability or partial acquisition as evidence for the unresolved clone-control question.

C EXPLORATORY FINGERPRINT AND SELECTION ANALYSES

Both analyses are post-hoc on the archived confirmatory cells (18 builder×seed cells, 384 admitted candidates, core-400 outcomes). The versioned `experiment/revision/` scripts now use the same canonical loader, frozen task list, membership and source checks as the main reanalysis. They

k	Selection	Oracle (%)	Best (%)	H (pp)	Dev-fixed (%)
4	top-acc	73.98	66.64	7.34	65.10
4	random	73.90	66.03	7.87	64.66
4	div-only	74.22	66.18	8.04	65.13
8	top-acc	76.03	67.11	8.93	65.10
8	random	76.15	66.76	9.39	64.95
8	div-only	76.39	66.68	9.70	65.10

Table 8: Canonical descriptive W1 reanalysis. Every pool contains k selected candidates plus the common bare. Means over 18 cells and 100 overlapping database splits; no confidence intervals. Dev-fixed is chosen only on development data; Best uses held-out outcomes.

remain descriptive: these tasks had already been analyzed, and this is not independent transfer validation.

Rich fingerprints. To isolate the loader change, the W3 reanalysis retains the legacy lossy SQL text transform: semicolon deletion, backtick replacement, lowercasing even segments after splitting on single quotes, and whitespace joining. This is not a SQL parser or semantic normalization; it can alter literals and quoted identifiers. Among 364,428 pair-task observations with two identical, nonempty transformed SQL strings, 0 have different verdicts. This is an observed consistency check, not a property guaranteed by construction.

Pair-level SQL dissimilarity averages (1-token-Jaccard) only over verdict-agreeing tasks with two nonempty SQL strings. Its Spearman correlation with overall outcome disagreement is 0.597 across 4,386 within-cell pairs and 0.588 across 1,010 within-arm pairs. Pairs share candidates and tasks, and the SQL metric conditions on verdict agreement; these correlations have no independence-based significance or causal interpretation.

In R2’s 70-harness first cache-off acquisition, call counts change on 506/28000 comparisons and verdicts on 2391/28000. Transformed SQL changes on 19569/27936 comparisons (70.05%); 64 comparisons lack two nonempty SQL strings and are excluded only from the SQL denominator. The mean token Jaccard among changed SQL pairs is 0.619. These channel-specific rates do not define an effect-size noise floor. Canonical logical-call costs are reported separately in Table 7.

Diversity-aware selection. Each cell pools all admitted candidates across arms. The reanalysis retains the legacy seed, 100 random database-half splits, 200 random subset draws per split and cell, and candidate-selection rules. It records a new deterministic schedule: the historical set-derived cell order was not archived, so identical historical splits and draws cannot be recovered from the seed alone. Selection uses development databases; evaluation uses the complementary databases. The overlapping splits are not 1,800 independent experiments. Methods are top- k development accuracy, uniform random subsets, greedy development oracle coverage, and greedy mean accuracy plus λ times oracle coverage for $\lambda \in \{0.25, 0.5, 1, 2\}$. All candidate choices are frozen within a split before its evaluation.

Table 8 retains the common AD bare in every evaluated subset; k counts candidates only. The JSON also records candidate-only values to distinguish the historical estimand. Oracle and best-fixed use held-out outcomes as descriptive upper bounds. Dev-fixed instead chooses one member using development accuracy, with ascending member-name ties and bare first. Deterministic methods already retain a globally dev-best candidate; differences in this comparator may arise from which tied maxima survive, not improved maximum development accuracy.

At $k = 8$, div-only versus top-acc changes oracle by +0.35 pp, best-fixed by −0.42 pp, and headroom by +0.78 pp. Dev-fixed changes by −0.001 pp. These are distinct quantities, and none is an independently validated routing gain. We withdraw the legacy cell-only confidence intervals; the current table is descriptive while inference covering database/task sampling and re-selection remains outstanding. The earlier normalized-headroom ratio is retained only in the historical artifact and is not carried into the revised claims.

Re-selection does not establish interval coverage. An exact counterexample rules out a universal 95% coverage guarantee for our proposed positive-weight re-selection percentile interval. Let nine independent databases each contain one Bernoulli(p) task state. Sixteen fixed candidates comprise eight state-0 specialists followed by eight state-1 specialists; bare is always wrong. Apply the canonical $k = 8$ rules to four development and five held-out databases. With the recorded tie rules, the expected div-only minus top-acc headroom statistic is $(1 - p^4) \mathbb{E}[\min(B, 5 - B)/5] > 0$, where $B \sim \text{Binomial}(5, p)$. On the event that all nine states coincide, every positive reweighting produces zero headroom for both methods, so the percentile interval is $[0, 0]$. Coverage is therefore at most $1 - p^9 - (1 - p)^9$, or 61.26% at $p = 0.1$. Averaging identical cells or overlapping fixed splits does not remove this event. This bound concerns the stated construction and interval, not BIRD-specific coverage or the impossibility of other inference methods.

C.1 ROUTING AND EXPLORATORY SELECTION

The Phase-I leave-one-harness-out predictor did not beat the best fixed harness in the tested DeepSeek population (-3.4 pp); the Qwen estimate was $+0.9$ pp (CI $[-5.2, +6.9]$), and task-held-out repair AUROC was 0.512. These predictor and split-specific results do not establish value prediction. The canonical W1 analysis is likewise descriptive: at $k = 8$, diversity versus top-accuracy selection changes oracle by $+0.35$ pp, best-fixed by -0.42 pp, and headroom by $+0.78$ pp; the development-selected fixed member changes by -0.001 pp. None is a validated routing gain (Appendix C).

D PHASE-I ADDITIONAL RESULTS

Per-harness accuracies, per-fold LOHO tables, bootstrap intervals, fix-set audits, K-subsampling curves, the task-held-out control, and the second-target replication (Qwen3.8-Flash as frozen target: collapse replicates; the strategy-forced population retains headroom; the prompt-side human control does not transfer wholesale) are included in the anonymous code release with their generating scripts.

E MODELS AND VERSIONS

Table 9 lists every model used in this paper with its exact version string as sent to the API. All models were accessed through a single commercial OpenAI-compatible API aggregator (the same provider for every role); the BIRD data-collection window is 2026-08-30 through 2026-09-08. The frozen target is always called at temperature 0. Phase-II harness generation uses temperature 0.7 with a 16,384-token budget; Phase-I generation settings are recorded per tool in the code release. Phase I used a proxy execution judge; Phase II uses the official BIRD scorer (30 s timeout), with the legacy judge as a secondary audit. The MATH-500 probe uses the answer comparator in Appendix J; no LLM judge produces reported accuracy.

F WORKED EXAMPLES FROM THE PHASE-I TRACE AUDIT

To make the failure taxonomy of §4.1 concrete, this section walks one BIRD task through the audited harnesses. Artifacts are verbatim from the day-1 trace audit (5 harnesses $\times$ 10 tasks); its logger caps stored SQL excerpts at 120 characters, so truncations are marked *[trunc.]*. Full harness sources and traces are in the code release.

The task. BIRD dev-set question 352 (card_games database; question text verbatim, the typo is the dataset’s):

Calculate the percentage of the cards availabe in Chinese Simplified.

Its BIRD hint:

Role	Model (exact version string)	Used in
Frozen target solver	GLM-5.3-Flash (temperature 0)	Phases I–II
Second frozen target (replication)	Qwen3.8-Flash	Phase I
Phase-I builders	GLM-5.3-Flash, Qwen3.8-Flash, DeepSeek-V4-Flash-Vision-Exp	§A.1
Phase-II builders	GLM-5.3, Qwen3.8-Max, DeepSeek-V4-Pro, Kimi-K3, MiniMax-M3, ERNIE-5.0-Thinking-Preview	Phase II
Gate-instrument development	GPT-6	App. G
Control authoring	gpt-5.5 (v2/v3 sets), gpt-5.6-sol (v4 set)	App. G
Blinded control re-view/adjudication	gpt-5.6-sol (v2/v3), gpt-5.5 (v4, semantic adjudication)	App. G
Judge	proxy / official BIRD execution scorer; MATH answer comparator	see text

Table 9: Complete model inventory. The target is held frozen within each phase; Phase-II builders come from six independent labs.

```
'Chinese Simplified' is the language; percentage =
Divide(Sum(id where language = 'Chinese Simplified'),
Count(id)) *100
```

Gold SQL (executing the two on the released database yields 8.7734 and 8.7728 respectively — the official scorer separates them):

```
SELECT CAST(SUM(CASE WHEN T2.language = 'Chinese Simplified'
THEN 1 ELSE 0 END) AS REAL) * 100 / COUNT(T1.id) FROM cards
AS T1 INNER JOIN foreign_data AS T2 ON T1.uuid = T2.uuid
```

The bare harness answers:

```
SELECT SUM(CASE WHEN language = 'Chinese Simplified'
THEN 1 ELSE 0 END) * 100.0 / COUNT(id) AS percentage FROM
foreign_data;
```

marked wrong by the official scorer: it counts directly over `foreign_data`, whereas the gold query counts after joining to `cards`; the hint does not spell out that join.

react (T2: mechanism present but untriggered). `react`’s control flow does execute SQL and retries on database errors. But its first call is byte-identical to `bare`’s prompt, and here the generated SQL executes without error, so the error-triggered retry never fires and the final SQL equals `bare`’s. Across all 10 audited tasks the pattern held: SQL executions present, final SQL byte-identical to `bare`’s on 10/10 — the feedback loop is real code whose repair branch did not activate on these tasks. Controlled failure injection is needed to test whether feedback changes the output.

cand.bird.g1.b0r0.g0 (T1: mechanism described, not executed). This candidate’s docstring claims a “conservative repair”: cap the solver’s maximum output tokens and demand an answer-only block. Its `solve()` is structurally `bare`’s: one LLM call, then `extract_sql`. The token cap exists only as an `os.environ.setdefault` default (2,048 against the runtime’s 32,000) — no harness control flow implements it, and it never binds on these tasks because a SQL answer is orders of magnitude below the cap. On the worked task it issues exactly one call, executes nothing, and returns the same wrong query as `bare` modulo whitespace (counting over `foreign_data` without the join). Across the 10 audited tasks it records zero SQL executions on all 10, and its final SQL is byte-identical to `bare`’s on 6 (the remaining differences trace to the reworded system prompt, not to any mechanism). The most common Phase-I failure class is therefore not broken machinery but absent machinery: the claimed mechanism exists only in prose.

cand.bird.g1.b0r1.g0 (code-level diversity, behavioral identity). This candidate is a genuinely different program: it wraps the single coder call in a daemon-thread watchdog and fires one hedged duplicate re-ask if the call stalls past 40 s. The mechanism exists in the control flow — but its trigger (a stalled API call) never occurred in the audit: zero SQL executions, and the final SQL equaled bare’s on all 10 audited tasks. Code difference without behavioral difference survives even when the added mechanism is real: it simply never fires.

cand.bird.g1.b1r1.g1 (T3: a broken multi-turn path). The one audited candidate implementing genuine execute-inspect-repair returns the model’s *prose* as final SQL. On a task bare answered correctly (“State the alternative languages available for card named Annul numbered 29.”), its logged final SQL begins, verbatim:

```
with "only a single read-only SELECT is allowed". The
previous "query" was actually just prose text (the model
wrote rea[trunc.]
```

The repair re-prompt invites the failure: asked to fix a query that “was actually just prose”, the frozen target responds with more prose reasoning, no SQL block is found, and the reasoning text is submitted as the query. The item flips from bare-correct to harness-wrong.

G INDEPENDENT CONSTRUCTION OF INSTRUMENT CONTROLS

We tested the post-review Gate-v2 development instrument with 32 newly authored controls, separate from the controls used to develop its probes. The instrument was fixed before a separate model (gpt-5.5; Table 9) received the public contracts and callable interface, without its implementation or earlier controls. The author supplied two intended conforming and two intended nonconforming implementations for each of the eight mechanism types. Control source, author-reference labels, instrument, measurement code, and execution order were hashed before the first evaluation. No benchmark or provider generation was run, and this first measurement is retained unchanged.

Declared mechanism	Structural pass	Structural fail	Review required
Repair	2	2	0
Execution-result vote	2	2	0
Schema linking	0	4	0
Hint guard	0	0	4
Two views	0	3	1
Decomposition	1	3	0
Error classification	0	4	0
Format guard	0	0	4
Total	5	18	9

Table 10: First evaluation of 32 independently authored controls by the fixed development instrument. No case had an execution failure. These are instrument states, not correctness labels or population error rates; the author’s intended positive/negative labels require separate scrutiny.

The nine review-required cases were retained. A third model (gpt-5.6-sol) received the full control source and traces without author-reference labels or automatic verdicts, for source-contract review of all 32 controls and semantic adjudication of the four designated mechanism types. Independent source review supported conformance for one decomposition control and two error-classification controls that the instrument rejected. The decomposition stub returned the payload of the first plan step whose text occurred in a request; retaining the earlier step in the context of the next request therefore retrieved the earlier payload again. The classification check removed the exact error text but still counted an ordinary `Schema :` heading as a category, making the extracted category set disagree with the expected error class. These are instrument failures on these controls, not observed failures of a target model. Three author-intended positive controls were disputed: two schema-linking controls retained an unrelated table, and one two-view control batched its requests and mishandled an empty result. Their original labels and the review disagreements were retained separately. Among the 29 controls whose author labels and source review agreed, the instrument

rejected 3 of 13 conforming controls; all 16 nonconforming controls were rejected after the designated semantic review. The instrument therefore did not pass this finite control check. These counts do not estimate error rates in a generated population. Semantic adjudication of the designated 16 controls yielded 7 conforming, 8 nonconforming, and 1 indeterminate case. The indeterminate case already had a structural failure, which was retained without resolving its semantics. This small, deliberately balanced control set is not a random sample of generated harnesses. Passing it would not validate the production interface or replace the common-raw-pool comparison of admission policies. The source, frozen inputs, full traces, and adjudications are retained in the accompanying revision artifacts.

Revised instrument and a new independent control set. Gate-v3 corrected decomposition request attribution and treated category keywords and shared request templates as review clues rather than automatic rejections. We fixed this version before fresh model contexts (gpt-5.5 authoring, gpt-5.6-sol review) authored and reviewed a new set of 32 controls, again two intended positive and two intended negative controls per mechanism. The review packet withheld author labels, automatic verdicts, and instrument-derived attribution metadata. All 32 controls received source review; the 20 controls in the four prompt/formulation groups plus decomposition also received semantic or data-flow review. Structural outcomes were 8 passes, 10 failures, and 14 review-required cases, with no execution failures. Following one recorded evidence clarification, source review classified 15 controls as conforming and 17 as nonconforming; semantic review classified 8 as conforming, 11 as nonconforming, and 1 as indeterminate.

The indeterminate classification control requested an intermediate error category, but the synthetic interface returned SQL text. Its fallback then selected a semantics repair action. This establishes a response-type mismatch in the fixture, not a real solver’s classification error. A second classification control was reference-disputed: its source mapped the supplied schema-failure message to a semantics action. Among the 31 undisputed references, 14 of 15 conforming controls were accepted and one remained unresolved; all 16 nonconforming controls were rejected. The prespecified finite check therefore remained unmet because of the unresolved case and insufficient agreed positive coverage for error classification. One voting control’s conformance applied only to result arbitration; it did not establish task-conditioned candidate generation. These finite mechanism checks do not validate complete harness operation or replace the planned common-pool policy comparison.

Runtime response contracts. We subsequently tested seven archived ungated, strategy-forced candidates, selecting the first available filename per strategy rather than by performance; no format-guard candidate was available (Appendix B). The 27 synthetic scenarios exposed both candidate and instrument limitations. The two-view candidate skipped its second execution when the first result was nonempty, contrary to its original “execute both” declaration. In contrast, the schema-linking candidate requested JSON but received plain text, and the full plan in decomposition requests prevented the frozen probe from uniquely attributing the current target. We retained those observations and tested source-assigned response profiles in 11 further development scenarios. Dedicated subquestion fields identified the current target, executable SQL payloads reached the final assembly, and four errors produced by an in-memory SQLite database elicited the corresponding syntax or schema repair instructions. Schema-linking JSON changed the explicit table/column restrictions in the SQL-writing request, although the full schema remained present. Retaining schema context therefore cannot by itself establish that the linked subset was ignored. These exposed-candidate checks distinguish response-interface failures from observable candidate deviations; they do not replace independent calibration or provide revised benchmark outcomes.

Unified interface and premeasurement reference review. Gate-v4 combined the source-assigned response interfaces with the five unchanged Gate-v3 mechanism probes. A separately authored set of 32 controls (gpt-5.6-sol) underwent blinded source review before execution. The author was familiar with earlier control failures; schema-response and declared voting-parameter mismatches found in this source review were corrected, with the initial versions retained. The revised references had two agreed conforming and two agreed nonconforming controls per mechanism. After freezing their source, profiles, and dependencies, one measurement covered 124 scenarios without execution exceptions. Structural results were 4 passes, 6 failures, and 22 review-required cases.

Arm	K_{cand}	K_{total}	Bare (%)	Best (%)	Oracle (%)	H (pp)
A	5.78	6.78	63.25	66.00	74.51	8.51
B	6.11	7.11	63.25	66.12	74.39	8.26
C	5.61	6.61	63.25	65.64	75.01	9.38
D	3.83	4.83	63.25	65.65	73.90	8.25

Table 11: **Same-core headroom decomposition.** All arms use identical 400 tasks and identical bare outcomes. K_{cand} excludes bare and $K_{\text{total}} = K_{\text{cand}} + 1$; counts and accuracies are cell means. Headroom is oracle minus best-fixed accuracy, not realized routing gain.

A new review context (gpt-5.5) received all source and actual traces without reference labels, source-review decisions, or automatic verdicts. Its first judgments were 18 conforming and 14 non-conforming. Retaining structural failures in the prespecified hybrid rule accepted all 16 conforming references but also one nonconforming hint control, so the finite zero-error criterion was not met. That control correctly restated existing hints but invented a mandatory sort order when no hint was supplied. A recorded, targeted evidence clarification corrected this oversight and a two-view judgment that had treated two samples from one request as independent formulations. The latter control was already rejected structurally. The clarified decisions agreed with all 32 references, but this follow-up does not replace the first blinded result or constitute a new independent calibration pass. No instrument rerun, provider generation, or benchmark evaluation was performed in this check.

Scope of the declaration vocabulary. These controls cover the eight strategy-forced contracts, not the free arm’s four declarations (repair, vote, twostage, and plain). In the original free prompt, vote specifies multiple samples and selection but not three-way majority voting; twostage specifies artifact consumption without a particular response interface. The plain declaration also specifies no control-flow mechanism, whereas the free prompt requests a control-flow change. Applying the eight-strategy gate to these declarations therefore requires separately fixed semantic rules and an indeterminate policy, rather than silently treating vote as vote3 or choosing a compatible profile after observing its result. This source-level scope gap does not estimate false-rejection rates for generated candidates.

G.1 SAME-CORE HEADROOM DECOMPOSITION

H DIAGNOSTICS PIPELINE AND VALIDATION (THIS REVISION)

Five question classes. For a population of harnesses and a bare reference, the diagnostics pipeline evaluates five classes of claims on an explicit execution matrix $Y[h, x, r, c]$ (member, task, repeat, execution condition): **(A)** measurement sufficiency (manifest consistency, judge replay with an explicit executed/not-executed state, identity uniqueness; a check that was not run is reported as not-run and can never render the class SUPPORTED); **(B)** single-shot coverage (oracle minus best-fixed, with dev-fixed reported separately); **(C)** stability, in two separated estimands; **(D)** pre-execution selectability (a frozen one-call selection policy versus dev-fixed); **(E)** budget-matched policy utility under a frozen 1-call budget with $\lambda=1$ harm weighting (offline, one-call; E leaves INSUFFICIENT_EVIDENCE unless the per-task call budget is verified by per-record evidence or is part of the generating design). Each class returns SUPPORTED, REFUTED_WITHIN_SCOPE, INSUFFICIENT_EVIDENCE, or NOT_APPLICABLE with scope and minimal missing evidence; classes are never collapsed into a single score.

Two stability estimands. *C-rank*: pairwise same-condition accuracy differences with a SHARED task-resample bootstrap and max-T simultaneous 95% intervals over all $M(M-1)/2$ canonical member pairs (family-wise error control; the pre-v3 per-pair intervals did not adjust for 36 simultaneous comparisons), plus sign consistency across repeats; the pair direction is fixed by member identity, so population conclusions are invariant to member ordering. The population state means at least one pair differs after multiplicity adjustment, not that the complete ranking is stable. *C-comp*: for a general population stable complementarity is $H_{\text{stable}} = \mathbb{E} \max_h q_h(x) - \max_h \mathbb{E} q_h(x)$, zero exactly when one member is expectation-best on every task; for two members this reduces to the half-absolute-value form of Eq. (2), which is the only form displayed in the main text. Re-

peats with differing execution-condition fields or differing member source hashes are refused as matched repeats and force INSUFFICIENT EVIDENCE; byte-identical outcome matrices are refused unless both repeats carry distinct, explicitly independent execution provenance (legitimate near-deterministic programs can reproduce identical matrices, so identity equality triggers provenance verification rather than an automatic rejection). Pair-level imputation bounds over unknown cells are reported alongside the recorded success rate.

The v3 C-comp decision (post 2026-09-19 review). The repeat-averaged plug-in of H_{stable} is retained as a *descriptive* field only; it never determines a scientific state. Two known-truth counterexamples from the review are encoded as regression tests: all $q = 0.9$ with $M=9, T=400, R=3$ gives a plug-in $\hat{H} = 8.69$ pp (the mean over the 300 hash-bound simulated datasets), and the sealed same-code clone arm prints 2.67 pp — both with true $H_{\text{stable}} = 0$. The calibrated statistic is the cross-fitted frozen-selection gain G described in §3; the decision uses the real-minus-clone difference D .

What holds at the population level, and what is checked empirically. Let $\theta^* = G$ be the population gain under the frozen selection protocol S . On every task the selected member’s probability is at most $\max_h q_h(x)$, so $\theta^* \leq H_{\text{stable}}$. The null $H_{\text{stable}} = 0$ is the global-dominance cone: some h_0 satisfies $q_{h_0}(x) \geq q_h(x)$ for all h and almost every x . Under dominance a population-optimal frozen mapping routes to the dominant member, so the population gain is $\theta^* \leq 0$ throughout the cone, not just at equal ability. This is a statement about the *population* selector, not about the finite- (R, T) statistic: the discovery argmax is selected from noisy repeats, and a weaker member that ties the noisy discovery score can be chosen in a finite sample (the sealed clone arm prints $G_{\text{clone}} = 0.55$ pp precisely from this selection noise), so $\theta^* \leq 0$ does not by itself bound the finite-sample rejection probability. The two-arm Monte Carlo randomization reference is exact at the equal-ability boundary, where permuted labels are exchangeable and the recomputed statistic (with the +1 correction) is a draw from the same conditional distribution. Over the rest of the dominance cone, false support is verified *empirically* on every null DGP in the calibration grid below — including heterogeneous difficulty, heterogeneous dominance, the 0.5/0.9 configuration, missingness, and sticky within-task latent states matched between arms — rather than asserted as an analytic finite-sample result. A level-.05 procedure can false-support at level .05 on a finite grid; the defensible claim is the observed worst rate and its Wilson interval, not impossibility.

Tie-breaking. Member-label tie-breaking in the frozen argmax is iid $U(0, 10^{-9})$ jitter averaged over ten frozen tie realizations on the real-data report (each calibration dataset uses one draw: under the null the tie bias has mean zero); a deterministic lexicographic rule is not label-invariant and is rejected by simulation. Label-symmetry is asserted directly by a permutation-invariance regression test: the 10-draw tie-averaged G, D , and state agree up to Monte Carlo tolerance when the member order is permuted identically in both arms.

Randomization mechanics. Labels are permuted independently within every task AND every repeat (a single joint permutation across repeats preserves a genuine crossover mapping and does not null the statistic: verified in development), recomputing both G ’s, their difference, and the two selections on every draw; the +1 Monte Carlo correction gives an exact discrete level. A paired task bootstrap supplies the sampling interval; a $\leq 10\%$ incomplete-task coverage gate, extremal unknown-cell bounds, and a positive G_{real} sign guard complete the rule, and without a clone arm no SUPPORTED state is emitted.

Known-truth simulation calibration. Binary tensors are drawn from specified $M \times T$ probability matrices (experiment/diagnostics/calibration.sim.py): iid equal ability at constant q and with shared task difficulty; global dominance with a clean dominant fixed member, the reviewer’s 0.5/0.9 configuration, and heterogeneous dominance; and crossover alternatives of increasing strength over two latent task types, plus difficulty-related partial crossover. The grid spans $M \in \{3, 9, 20\}$, $T \in \{100, 400, 1000\}$, $R \in \{3, 5, 10, 20\}$, complete data, 5% MCAR, and 3% hardness-related missingness; the paper table focuses on the $(9, 400, \cdot)$ acquisition design with 300 Monte Carlo datasets per cell (150 in sweeps). Every cell reports the support, refutation, abstention and error rates of four procedures (plug-in; one-split frozen gain; repeat-cross-fitted gain; clone-calibrated v3 decision), interval coverage against the true parameter, and mean estimates. Results (artifacts under artifacts/diagnostics/calibration.v3/): the plug-in

false-supports at rate ≈ 1 on every null at all grid points; v3 holds the null support rate at or below the nominal level across nulls and missingness regimes, and its power on alternatives increases with R (small/moderate interactions remain unresolved at $R=3$, which is reported as an abstention property rather than hidden). The earlier eight mechanism-derived classes and the four repeat-based classes (C9–C12, with C9/C10 now paired with registered same-code clone arms) continue to match their hash-frozen expected states; invariant tests additionally assert permutation invariance, dominance $\neq$ complementarity, identity refusal, and the two review counterexamples directly.

Finite-repeat bias and the clone-null calibration. Writing $\hat{q}_h(x)$ for a repeat-averaged estimate, $\mathbb{E} \max_h \hat{q}_h(x) \geq \max_h \mathbb{E} \hat{q}_h(x)$ by Jensen for each task, with strict inequality when finite- R noise can change the per-task argmax. The plug-in *headroom* is a difference of two such maxima; Jensen bounds each term separately and therefore does NOT establish that the difference is upward biased in general — in a two-member, two-task crossing example the plug-in can be downward biased (true $H=0.40$, expected plug-in over three repeats ≈ 0.34). What is established is stronger and narrower: under same-code exchangeability the true H_{stable} is zero while the plug-in is positive with probability approaching one (selection noise on the per-task argmax; $\hat{H}=8.69$ pp in the $q=0.9$, $M=9$, $T=400$, $R=3$ known-truth simulation, and 2.67 pp descriptive on the sealed clone arm’s 383 complete tasks (the real panel prints 2.54 pp on a different 350-task complete set; the two are not a paired comparison)). The calibrated procedure therefore never decides from the plug-in: a discovery/validation cross-fit freezes the per-task member mapping on discovery repeats and scores it on held-out repeats against the best *fixed* member (so a globally dominant member is always the frozen pick), and the real-minus-clone difference removes the residual finite- (R, T) selection-noise component with a two-arm Monte Carlo randomization reference (+1 correction; exact at the equality boundary, empirically checked across the dominance grid rather than asserted analytically throughout the cone; Appendix H). The identity gates handle the degenerate case outright by requiring provenance-verified independent executions; identical matrices without distinct provenance are refused.

Class D and E formal definitions. Class D evaluates one frozen deterministic policy $\pi_Z : Z(x) \mapsto h$ — pre-execution features (question text, frozen mechanism labels, training-fold dev accuracies), fitted on development tasks only — by $D(\pi_Z) = \mathbb{E}_x[Y(x, \pi_Z(x)) - Y(x, h_{\text{dev-fixed}})]$, with an abstention rule falling back to bare when the top-2 selection-probability gap is below the frozen margin. Class E evaluates $U_b(\pi) = \mathbb{E}_x[\Delta(x, \pi(x))]$ under a uniform per-task call budget b with $\lambda=1$ harm weighting (one regression cancels one repair); it is admissible only when per-record evidence verifies the budget on every evaluated member-task pair, and abstains otherwise.

Validation on hand-built known-truth controls. The frozen package has two layers: the original eight mechanism-derived classes $\times 2$ instances (16 controls, expectations unchanged), plus four repeat-based classes $\times 2$ instances exercising the stability paths single-shot controls cannot reach — global dominance (C-rank SUPPORTED, C-comp REFUTED), equal-mean crossover skill (C-rank INSUFFICIENT, C-comp SUPPORTED under clone calibration, and the visible stratum makes D/E SUPPORTED), a condition-mixing trap, and a cloned-repeats trap (both must abstain); C9/C10 carry registered same-code clone arms as required by the v3 decision. Expected states are frozen in the hashed manifest before any blinded run and derive from the generating mechanisms. Invariant tests additionally assert permutation invariance, dominance $\neq$ complementarity, identity refusal, the two 2026-09-19 review counterexamples (iid noise and 0.5/0.9 dominance must not support; crossover must support with adequate repeats), and that the decision never supports without a clone arm. Calibration matched on all state cells; the blinded set (12 controls + 2 auditor challenges re-executed from the pre-committed, hash-bound spec under the current code) matched on all 14×6 cells: zero false supports, the positive control recognized, and the abstention cap holds. These discrete controls complement, but do not replace, the Monte Carlo false-support/power calibration of §3, which is what establishes the v3 test’s statistical size. The earlier stored challenge result whose CH2 numbers contradict its own spec (best-fixed 0.94 is unreachable under the generating rule, which forces 0.79) is retained as provenance only; spec-faithful re-execution matches the auditor’s pre-committed expectations on every cell.

Application to the archives. Applied to the MATH-500 archive (36 members $\times$ 400 tasks) and the 18-cell BIRD Phase-II archive. On BIRD, the integrity class (A) is REFUTED_WITHIN_SCOPE

for every cell: the frozen input manifests contain overlapping collection files whose rows duplicate on the canonical key (resolved by the archived first-write-wins policy, with recorded verdict-conflict counts), so the coverage numbers below are computed on that deduplicated view, not on a conflict-free archive; judge replay is not executed on this path and is reported as not-run, not passed. Subject to that scope, all cells show single-shot coverage (B SUPPORTED); stability, selectability, and deployability are INSUFFICIENT EVIDENCE by the frozen rules because the archives contain no matched repeats and no dev/eval split for policy training — not because costs are absent: per-record logical calls are archived and are read on the same canonical deduplicated view as the rest of the reanalysis (259,304 calls across 153,600 candidate-task core-subset records, Table 7), though token/dollar costs remain unrecoverable. On MATH the frozen selection policy, trained on the first 80 of the 400 eval tasks held out of all D/E claims (the archive contains no dev-set executions: the 100-task dev split of Appendix J was used only for generation smoke tests, so the policy split is carved from the eval side, leaving 320 scored tasks), ties with dev-fixed (bare) on every eval task, so selectability and deployability are INSUFFICIENT EVIDENCE rather than refuted. Three published numbers were corrected (membership-basis of the outcome-vector count, now 33/36 members including bare and 32/35 generated harnesses; the generated-harness accuracy range 0.6825–0.915, previously printed as 0.68–0.87; the headroom decomposition with best-fixed = bare stated explicitly), two overloaded metric names were disambiguated (union-all-fail vs any-candidate-fail), and one preserved-judgment control confirms a reported pair (union repair 76.0% / union-all-fail 0.8%) under the explicit definition. A hash-bound re-execution of the versioned endpoint-sensitive judge over all 14,400 MATH verdicts changes 14 of them — all v1 false rejects of interval-endpoint answers on one task (math500_split#103); the v1 replay itself is internally consistent, and an earlier stored 0-flip artifact is not reproducible from the committed code and is superseded. The bracket-merging defect the new judge was designed to catch therefore does fire in this archive, on exactly one task.

I WP-1R/WP-2R FRESH COLLECTION: PROTOCOL AND EXECUTION RECORD

Frozen protocol. The acquisition protocol (review-stage/REAL_EVIDENCE_PROTOCOL_V1.md, v1.0 frozen 2026-09-15, with dated amendments: v1.1 (statistical and budget fixes) and v1.2 (operational test specification) preceded any confirmatory outcome; v1.3 (attempt ceiling 45,000→60,000, matching the task authorization) and the concurrency amendment (2→6) were made mid-collection for throughput and are disclosed as condition-relevant history) fixes: the panel draw (eligibility by frozen repeat-0 mean logical calls ≤ 3.05 — a budget-feasibility filter disclosed as biasing the panel toward low-call mechanisms, not a uniform sample of the 35 generated members; then a salted-hash draw, one stratum seat per builder plus two at-large seats, salt wplr-panel-v1-20260915, recorded in WP1R_PANEL_DRAW.json); $R=3$ fresh repeats per arm; the nine-slot same-code clone arm (clone-cl..c9, byte-identical bare source) as the exchangeable equal-ability reference; the 400 eval and 100 dev tasks of the frozen MATH-500 split (SHA256 recorded); eval tasks were analyzed before in the one-run mini-audit, so results are labeled retrospective-task/prospective-execution, not unseen-task validation.

Statistical procedure (v3, post 2026-09-19 review). The primary complementarity statistic is the cross-fitted frozen-selection gain G of §3: in each of three folds, two repeats freeze the per-task member mapping and the best *fixed* member, the third repeat scores the frozen mapping against that fixed member; G is the per-task mean averaged over folds. Tie-breaking is label-symmetric: because $R=2$ discovery scores lie on $\{0, \frac{1}{2}, 1\}$ and ties dominate at high accuracy, a deterministic lexicographic rule would bias the selection toward the first listed member, so iid $U(0, 10^{-9})$ jitter is drawn per (fold, task) and averaged over ten frozen tie realizations; a simulation over dominant-member positions verifies the statistic is order-independent. The decision statistic is the real-minus-clone difference $D = G_{\text{real}} - G_{\text{clone}}$ on the task intersection complete in both arms. Its reference distribution uses 10,000 two-arm randomizations: within every task and repeat, member labels are permuted independently in each arm and D is recomputed (a single permutation joint across repeats was rejected in development because it preserves a genuine crossover mapping and does not null it). A paired task bootstrap (2,000 draws, seed 20260915) gives the 95% interval. The frozen rule emits SUPPORTED only when the coverage gate passes ($\leq 10\%$ incomplete tasks in either arm; ≥ 30 paired tasks), the randomization $p < 0.05$, the bootstrap lower bound is positive, $D > 1.0$ pp,

and G_{real} 's own lower bound is positive; REFUTED requires the D upper bound within the margin and a small equivalence p ; otherwise INSUFFICIENT. The acceptance rule that any of these states terminates the study without re-running remains. The earlier v2 procedure — a discovery-repeat argmax scored against bare on repeats 2–3 (A8.6) — is retained as a labeled secondary statistic, not a complementarity test (under pure global dominance with bare at 0.5 and the other members at 0.9 its expectation is $\approx 0.5(1 - 0.1^8) \cdot 0.4 \approx 0.20$). Two frozen sensitivity branches accompany it (within-task available-repeat mean imputation; exclusion of doubly-executed conflict cells).

Execution record and continuation merge. Collection ran 2026-09-15 through 2026-09-19 under a 60,000-attempt global budget with per-cell atomic reservation. Two provider-timeout stops (2026-09-17, 2026-09-18) split the real arm across three continuation ledgers and the clone arm across two; each continuation re-binds the identical frozen solver condition (verified field-by-field at merge). The merge rule is deterministic: a completed observation beats an unknown_remote marker; among completed records of the same cell the earliest ledger wins. Across the real arm's 19,318 rows it collapsed 8,418 migration copies, preferred a completed record over an unknown_remote marker 14 times, completed an earlier unknown marker 9 times, and found 5 cells with two completed but disagreeing executions (first-completed retained; a frozen sensitivity branch excludes them instead). The clone arm's 14,967 rows collapsed to 10,800 unique cells (4,160 copies, 0 conflicts). 72 cells remain unknown_remote across all ledgers and stay NaN in the outcome matrices — never imputed — so the v3 coverage gate excludes their tasks (point G is a complete-case statistic, exact on the included tasks), and a scenario range for G under all-0 and all-1 filling of the excluded tasks' unknown outcomes (the tight fill-extremes, each recomputing the full cross-fit including selection changes) is reported as missing_extremal_G.bounds. The merged arms are hash-sealed (SEALED_eval_real.json, SEALED_eval_clone.json, SEALED_dev_real.json). Three collector source versions served the real arm across continuations (concurrency 2/2/6); each seal records the arm's bound collector hash, and the identity gates exclude run-identity fields (acquisition id) while this appendix discloses the concurrency and amendment history. The analysis code, calibration simulation, and their unit tests are in experiment/diagnostics/ccomp.v3.py, experiment/diagnostics/calibration.sim.py, and experiment/revision/wplr.analysis.py.

Budget semantics and realized accounting (E4). Three distinct comparisons are reported and never conflated. (i) *Equal-slot collection*: both arms execute nine members/slots for three repeats; this is the randomization design and does not imply matched spend. (ii) *Common per-task execution budget* (§4.3): offline comparison at $b \in \{3, 9, 15, 27\}$ executions per task between the dev-ranked panel portfolio and b of the 27 same-code executions, under post-execution *oracle* aggregation only; panel-minus-same-code oracle differences are +0.37, +0.03, −0.11, −0.25 pp. No cross-arm majority-vote accuracy is reported because the arms emit incompatible answer formats (boxed numbers versus free-form prose), so a modal answer key would be unfair. (iii) *Realized spend*: per-request accounting layers experiment-run > harness-execution > logical-call > provider-attempt; migration copies are excluded. The real arm made 22,192 logical calls (32,097 provider attempts; 54.17 M known tokens; 54.5 calls per eval task) versus the clone arm's 10,800 logical calls (10,804 attempts; 14.51 M tokens; exactly 27 per task) — a $2.05\times$ call and $3.73\times$ token ratio driven by the multi-call panel members; the conservative ceiling at the frozen break-even price is 1,803.82 vs 483.15 CNY (pricing not independently verifiable; reported as token counts times the frozen 33.3 CNY per 1M tokens, never as billed amounts). Totals across all arms: 38,458 logical calls, 50,817 attempts, 79.8 M known tokens, 2,656.22 CNY ceiling. We therefore make no realized-cost-matched claim; the common-budget table is the cost-controlled comparison.

Selector spec (E3). π_Z is multinomial logistic regression over TF-IDF question features (max 300) and per-member dev accuracies computed within training folds only; 5-fold task-level CV on the 100 dev tasks with same-task repeats never split across folds; fold-model averaging; ties broken by member id; abstain to bare when the top-2 probability gap is below 0.15. Repeat aggregation uses the available-repeat mean per member-task (same-task repeats stay in one fold). Comparators: bare, dev-fixed, pre-fixed random member (seeded), M1b (dev-fixed with bare fallback on missing cells). E_1 (budget by construction) is not applicable because panel members make more than one call; E_b with $b=3$ calls per task is reported as a separately named estimand. Leakage controls: no answer, correctness, eval-derived difficulty, post-execution trace, or judge feature is used.

Policy	Eval acc. (%)	Δ vs dev-fixed (pp) [95% CI]
bare (fixed)	94.96	0.00 [0.00, 0.00]
dev-fixed	94.96	—
random (pre-fixed)	85.14	-10.12 [-13.27, -7.22]
M1b (dev-fixed + bare fallback)	94.96	0.00 [0.00, 0.00]
π_Z (frozen selector)	94.96	0.00 [0.00, 0.00]

Table 12: **WP-2R frozen comparator set on the 400 MATH-500 eval tasks** (repeat-mean correctness over three fresh repeats; retrospective tasks, prospective execution — no unseen-task claim). Paired differences are per-task against dev-fixed with task-level bootstrap CIs.

J CROSS-DOMAIN REPLICATION: MATH-500 MINI-AUDIT

To test whether the Phase-II clean-protocol finding (free ungated generation produces distinct one-run correctness vectors) is text-to-SQL-specific, we replicated the free-arm protocol on competition math.

Protocol. Six builders (the same models as Phase II), one seed, eight slots each, an $R=3$ raw-attempt cap per slot, neutral validity gate (mechanism-blind: parse, import, interface, non-empty answer on two dev questions). Frozen target GLM-5.3-Flash (same as the SQL study), temperature 0. MATH-500 (Lightman et al., 2023): 500 tasks frozen before generation (100 dev / 400 eval, seed 20260908). The 100 dev tasks are used only for generation-side smoke tests and are never executed in the evaluation archive; the selection-policy experiments of Appendix H therefore carve their training split (80 tasks) from the eval side, leaving 320 scored tasks for D/E claims. Judge: numeric equality where both sides parse as numbers (including a/b , $\frac{a}{b}$, decimals), else normalized LaTeX string equality (units, base subscripts, $^\circ$ stripped; $\text{text}\{X\}$ content preserved; $\text{pmatrix} \rightarrow \text{tuple}$). Judge self-consistency: 400/400 gold answers judge correct against themselves. bare accuracy 0.9375 under the matched answer-format instruction. Yield: 35/48 slots admitted (R_{artifact} 0.10–1.00; the two reasoning-heavy builders, GLM-5.3 and Qwen3.8-Max, exhausted the 32k-token generation budget on reasoning before emitting code in most slots — reported as-is). This constraint effectively removes two of the six builders from the evaluated pool, so the free-arm population under-represents reasoning-heavy builder styles; we do not adjust the budget post hoc, and the limitation is part of the reported protocol. This is a replication probe, not a factorial: no strategy forcing, no conformance gate.

Results (35 harnesses, 400 eval tasks). Mean pairwise outcome disagreement 10.6% (max 33.0%); 33 of the 36 members had unique outcome vectors when bare is counted (32 of the 35 generated harnesses); outcome-identical pairs 0.7% — no trace of Phase-I-style collapse. Union repair 76.0% of bare errors; the union of all generated candidates fails on 0.8% of bare-correct tasks (a *union-all-fail* rate, not an any-candidate or router harm). Observed oracle coverage exceeds bare by 4.75 pp, and the best fixed member is *bare itself* (0.9375; generated harnesses 0.6825–0.915); every generated member is worse individually. The one-run mini-audit originally did not collect repeated-execution or same-code clone controls; that gap is closed on this same MATH-500 task set by the fresh pre-registered WP-1R collection (§4.3), which executes an eight-member panel and nine same-code clone slots for three independent repeats and therefore does evaluate the stability estimands cross-domain. A hash-bound re-execution of the versioned endpoint-sensitive rescore of all 14,400 archived verdicts changes 14 of them (0.097%): all are v1 false rejects of interval-endpoint answers on a single task, so per-member accuracies rise by at most one task and no qualitative conclusion moves (Appendix H). An earlier stored artifact reporting zero flips is not reproducible from the committed code and is retained only as provenance.

Two failure-side observations. First, the clean-protocol diversity result replicates in a second domain with a different output space. Second, a quarantined pilot round (disclosed in PROTOCOL_CHANGELOG.md, entry G1) supplies an exploratory comparison: when the same six builders were (mistakenly) given a GSM8K-style task description whose numeric answer format did not match MATH-500, the generated population remained behaviorally diverse (16.2% disagreement, 19 unique outcome vectors) while its union repair collapsed to 18% — and code-level inspec-

Domain / analysis	n_H	n_{eval}	Disagree.	Oracle	Best-fixed	H (pp)
BIRD, A / D	18 cells	1,169	—	76.14 / 75.70%	67.58 / 67.30%	8.56 / 8.40
BIRD, D—A	18 paired	1,169	—	—	—	−0.16
MATH-500, free	35	400	10.6%	98.50%	93.75% (bare)	4.75

Table 13: Parallel domain evidence for the same measurement question. BIRD is the archived factorial analysis; MATH-500 is a one-seed exploratory free-arm probe. Designs are not pooled. Headroom is a one-run diagnostic, not selector accuracy or stable complementarity.

tion of the catastrophic members reproduced the Phase-I trace taxonomy verbatim (a generate-then-verify harness whose verifier rejected nearly every answer and whose fallback returned the stale loop variable or “0”). Diversity without utility, induced by a prompt-side protocol error invisible to code-level checks: the measurement mandate, not any code-level audit, is what detects it.

Scope. One seed, one target, no factorial, exploratory; the Phase-I collapse-inducing SQL protocol was not re-run in the math domain, so this replication speaks to the clean-protocol result only, not to the boundary of collapse.

J.1 PARALLEL DOMAIN EVIDENCE FOR THE MEASUREMENT BOUNDARY

Table 13 places the two domains side by side for the same diagnostic question, without pooling them. MATH-500 reproduces the boundary: one-run disagreement and 4.75 pp headroom coexist with bare as the best fixed member (93.75%). The one-run probe itself has no repaired-gate control; clone, repeat, and selector controls come from the separate fresh WP-1R collection (§4.3).